

Black Belt 6Sigma

1-way and 2-way ANOVA
Design of Experiments

Categorical variables: factors

In experiments we often want to gather several measurements for each of a few well-chosen **fixed points**

A **categorical variable** or a **factor** is a variable of nominal type which can assume a limited number of fixed values called the **levels** of the factor. These levels have no numerical meaning or ordinal relation.

e.g. factor 'Material' with 3 levels 'Aluminium', 'Zinc', 'Tungsten'

factor 'Age' with 4 levels '0–18', '19–35', '36–60', '>60'

One-factor experiments:

- ▶ Each factor level represents a certain population. The measurements corresponding to a level, or **replicates**, form a sample from that population.
- ▶ The group of measurements belonging to a factor level is called a **category**, **group**, **treatment** or **cell**
- ▶ If each group contains the same number of measurements, the experiment is **balanced** (we will always assume it is)

Comparing means of groups

Consider the factor A with a levels and n replicates per group

Assume all populations are **normal** with **different means**, but the **same variance**:

Let Y_{ij} be the j th measurement for the i th level of A, then

- ▶ The **means model** is

$$Y_{ij} = \mu_i + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim \mathbf{N}(0, \sigma^2) \text{ and independent}$$

- ▶ The **effects model** is

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim \mathbf{N}(0, \sigma^2) \text{ and independent, } \sum_{i=1}^a \tau_i = 0$$

for $i = 1, \dots, a$ and $j = 1, \dots, n$. So either we characterise the populations directly by their means μ_i , or by their 'effect' τ_i compared to the overall grand mean μ . All populations have common variance σ^2 .

Both models are equivalent of course and have $a + 1$ free parameters to estimate

- The goal is to test whether **factor A has an influence on the measurements** in a statistically significant way.

In other words: are the μ_i unequal, or are any of the τ_i nonzero?

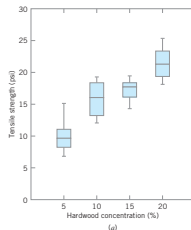
Comparing means of groups

For example, to investigate the presence of an **influence** of hardwood concentration in paper on its tensile strength, we fix $a = 4$ levels of the factor HardwoodConcentration and collect $n = 6$ measurements in each group:

■ TABLE 4.4

Tensile Strength of Paper (psi)

Hardwood Concentration (%)	Observations						Totals	Averages
	1	2	3	4	5	6		
5	7	8	15	11	9	10	60	10.00
10	12	17	13	18	19	15	94	15.67
15	14	18	19	17	16	18	102	17.00
20	19	25	22	23	18	20	127	21.17
							383	15.96

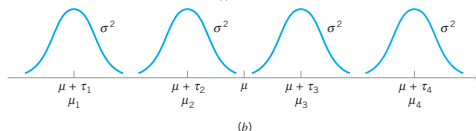


Hypothesis:

$$H_0: \mu_1 = \dots = \mu_a$$

$$\text{or } \tau_1 = \dots = \tau_a = 0$$

H_A : otherwise



■ FIGURE 4.11 (a) Box plots of hardwood concentration data. (b) Display of the model in equation 4.65 for the completely randomized single-factor experiment.

► If there were only 2 groups, we could use a two-sample t -test

► For $a > 2$, we can use the **one-way ANOVA** test to decide whether the factor does (H_A) or does not (H_0) have a significant influence on the response

Linear model for one factor: dummy variables

In fact, both the effects and the means model for a factor are special cases of a multiple linear regression model.

- Take e.g. a factor Color with n measurements at $a = 4$ color levels:

level	observations	
red	$Y_{11}, Y_{12}, \dots, Y_{1n}$	$\bar{Y}_{1.}$
green	$Y_{21}, Y_{22}, \dots, Y_{2n}$	$\bar{Y}_{2.}$
blue	$\dots$	
yellow	$Y_{a1}, Y_{a2}, \dots, Y_{an}$	$\bar{Y}_{a.}$
		$\bar{Y}_{..}$

group means:

$$\bar{Y}_{i.} = \frac{1}{n} \sum_{j=1}^n Y_{ij}, \quad i = 1, \dots, a$$

grand mean:

$$\bar{Y}_{..} = \frac{1}{a} \sum_{i=1}^a \bar{Y}_{i.} = \frac{1}{an} \sum_{i=1}^a \sum_{j=1}^n Y_{ij}$$

This needs to be fitted to the means model:

$$Y_{ij} = \mu_i + \varepsilon_{ij} \text{ with } \varepsilon_{ij} \text{ iid } \mathcal{N}(0, \sigma^2), \quad i = 1, \dots, a, \quad j = 1, \dots, n$$

→ the parameters μ_i and σ^2 can be estimated using ordinary least-squares (OLS) estimation, which R/Excel/SPSS/minitab/... can do.

Linear model for one factor: dummy variables

Analysis of variance

The OLS formulas for SS partitions also reduce to those for the means model:

► $SS_{\text{Total}} = (\mathbf{Y} - \bar{Y}\mathbf{1})^T(\mathbf{Y} - \bar{Y}\mathbf{1}) = \sum_{i=1}^a \sum_{j=1}^n (Y_{ij} - \bar{Y}_{..})^2$
#obs - 1 df = $an - 1$ df

► $SS_{\text{Error}} = \mathbf{e}^T \mathbf{e} = (\mathbf{Y} - \hat{\mathbf{Y}})^T(\mathbf{Y} - \hat{\mathbf{Y}}) = \sum_{i=1}^a \sum_{j=1}^n (Y_{ij} - \bar{Y}_{i.})^2$
#obs = an , $p = a$, so #obs - $p = a(n-1)$ df

→ unexplained SS: within-group variability

► $SS_{\text{Treatment}} = SS_R = (\hat{\mathbf{Y}} - \bar{Y}\mathbf{1})^T(\hat{\mathbf{Y}} - \bar{Y}\mathbf{1}) = n \sum_{i=1}^a (\bar{Y}_{i.} - \bar{Y}_{..})^2$
 $p = a$, so $p - 1 = a - 1$ df

→ explained SS: between-group variability

→ the SS partitioning still holds: $SS_{\text{Total}} = SS_{\text{Treatment}} + SS_{\text{Error}}$

One-way ANOVA test

Hypothesis

$$H_0: \tau_1 = \dots = \tau_a = 0 \quad \text{or: } \mu_1 = \dots = \mu_a$$

$$H_A: \exists i : \tau_i \neq 0, 1 \leq i \leq a \quad \text{or: group means are different}$$

Test statistic

under H_0 : as we know from OLS regression:

$$\frac{1}{\sigma^2} SS_{\text{Treatment}} \sim \text{Chi}^2_{a-1} \quad \text{and} \quad \frac{1}{\sigma^2} SS_{\text{Error}} \sim \text{Chi}^2_{a(n-1)} \quad \text{are independent}$$

Therefore, we can use an F -statistic to compare:

$$F = \frac{\frac{SS_{\text{Treatment}}}{\sigma^2(a-1)}}{\frac{SS_E}{\sigma^2 a(n-1)}} = \frac{MS_{\text{Treatment}}}{MS_{\text{Error}}} \sim F_{a-1, a(n-1)}$$

► Reject H_0 when $F > F_{1-\alpha, a-1, a(n-1)}$

One-way ANOVA test

We can also make the ANOVA table:

■ **TABLE 4.6**

The Analysis of Variance for a Single-Factor Experiment

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0
Treatments	$SS_{\text{Treatments}}$	$a - 1$	$MS_{\text{Treatments}}$	$\frac{MS_{\text{Treatments}}}{MS_E}$
Error	SS_E	$a(n - 1)$	MS_E	
Total	SS_T	$an - 1$		

Intuitively:

- Note that under H_0 , $E[MS_{\text{Error}}] = \sigma^2$ and

$$E[MS_{\text{Treatment}}] = \frac{E[SS_{\text{Treatment}}]}{a - 1} = \sigma^2 + \frac{n}{a - 1} \sum_{i=1}^a \tau_i^2 = \sigma^2$$

So numerator and denominator of the F -statistic are 'about equal', giving an F -value **close to 1**

- Under H_A , there *is* an effect and $E[MS_{\text{Treatment}}] > E[MS_{\text{Error}}]$
so we will typically see F -values (much) **larger than 1**

Example

EXAMPLE 4.12 The Paper Tensile Strength Experiment

Consider the paper tensile strength experiment described in Section 4.5.1. Use the analysis of variance to test the hypothe-

sis that different hardwood concentrations do not affect the mean tensile strength of the paper.

SOLUTION

The hypotheses are

$$H_0: \tau_1 = \tau_2 = \tau_3 = \tau_4 = 0$$

$$H_1: \tau_i \neq 0 \text{ for at least one } i$$

We will use $\alpha = 0.01$. The sums of squares for the ANOVA are computed from equations 4.73, 4.74, and 4.75 as follows:

$$\begin{aligned} SS_T &= \sum_{i=1}^4 \sum_{j=1}^6 y_{ij}^2 - \frac{y_{..}^2}{N} \\ &= (7)^2 + (8)^2 + \dots + (20)^2 - \frac{(383)^2}{24} = 512.96 \\ SS_{\text{Treatments}} &= \sum_{i=1}^4 \frac{y_{i.}^2}{n} - \frac{y_{..}^2}{N} \\ &= \frac{(60)^2 + (94)^2 + (102)^2 + (127)^2}{6} - \frac{(383)^2}{24} = 382.79 \\ SS_E &= SS_T - SS_{\text{Treatments}} \\ &= 512.96 - 382.79 = 130.17 \end{aligned}$$

We usually do not perform these calculations by hand. The ANOVA from Minitab is presented in Table 4.7. Since $F_{0.01,3,20} = 4.94$, we reject H_0 and conclude that hardwood concentration in the pulp significantly affects the strength of the paper. Note that the computer output reports a P -value for the test statistic $F = 19.61$ in Table 4.7 of zero. This is a truncated value. Appendix Table V reports that $F_{0.01,3,20} = 4.94$,

so clearly the P -value is smaller than 0.01. The actual P -value is $P = 3.59 \times 10^{-6}$. However, since the P -value is considerably smaller than $\alpha = 0.01$, we have strong evidence to conclude that H_0 is not true. Note that Minitab also provides some summary information about each level of hardwood concentration, including the confidence interval on each mean.

Example

■ TABLE 4.7

Minitab Analysis of Variance Output for the Paper Tensile Strength Experiment

One-Way Analysis of Variance					
Analysis of Variance					
Source	DF	SS	MS	F	P
Factor	3	382.79	127.60	19.61	0.000
Error	20	130.17	6.51		
Total	23	512.96			
Individual 95% CIs For Mean Based on Pooled StDev					
Level	N	Mean	StDev	—+——+——+——+	
5	6	10.000	2.828	(*—)	
10	6	15.667	2.805	(*—)	
15	6	17.000	1.789	(*—)	
20	6	21.167	2.639	(*—)	
Pooled StDev = 2.551				—+——+——+——+	
				10.0	15.0 20.0 25.0

Checking assumptions: residuals analysis

The **residuals** are the values $e_{ij} = y_{ij} - \bar{y}_i$. ($i = 1, \dots, a$ and $j = 1, \dots, n$)
= difference between observations and corresponding **treatment means**
= difference between observations and **the model's prediction**
→ used as a proxy for the **errors** ε_{ij} (!)

We check visually:

- ▶ Do the residuals have **constant variance**? Patterns?
- ▶ Are the residuals **normally distributed**?

Make a **normal probability plot**, plot of residuals **vs. factor levels** or **vs. treatment means**, etc.

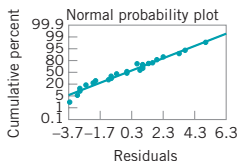


FIGURE 4.13
Normal probability plot of residuals from the hardwood concentration experiment.

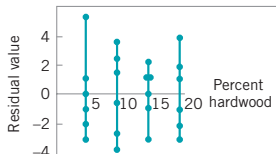


FIGURE 4.14 Plot of residuals versus factor levels.

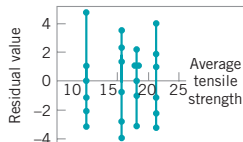
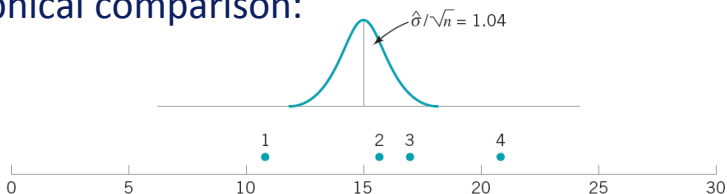


FIGURE 4.15 Plot of residuals versus $\bar{y}_i$.

Which means differ?

Graphical comparison:



■ **FIGURE 4.12** Tensile strength averages from the hardwood concentration experiment in relation to a normal distribution with standard deviation $\sqrt{MS_E/n} = \sqrt{6.51/6} = 1.04$



20% hardwood produces higher mean strength than the others; 5% hardwood produces lower strength; 10% and 15% hardwood don't differ but give lower strength than 20%.

More formal methods: e.g. Fisher's LSD

An Analysis of Data Example

- Consider an investigation into the formulation of a new “synthetic” fiber that will be used to make cloth for shirts. The response variable is tensile strength.
- The experimenter wants to determine the “best” level of cotton (in wt %) to combine with the synthetics.
- Cotton content can vary between 10 – 40 wt %, some non-linearity in the response is anticipated.
- The experimenter chooses 5 **levels** of cotton “content” 15, 20, 25, 30, and 35 wt %.
- The experiment is **replicated** 5 times – runs made in random order.

An Analysis of Data Example

Table 3-1 Data (in lb/in²) from the Tensile Strength Experiment

Cotton Weight Percentage	Observations					Total	Average
	1	2	3	4	5		
15	7	7	15	11	9	49	9.8
20	12	17	12	18	18	77	15.4
25	14	18	18	19	19	88	17.6
30	19	25	22	19	23	108	21.6
35	7	10	11	15	11	54	10.8
						376	15.04

- Does **changing** the cotton weight percent changes the mean tensile strength?
- Is there an **optimum** level for cotton content?

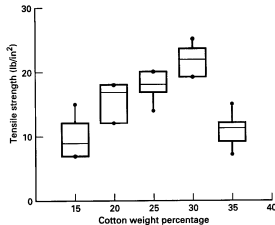


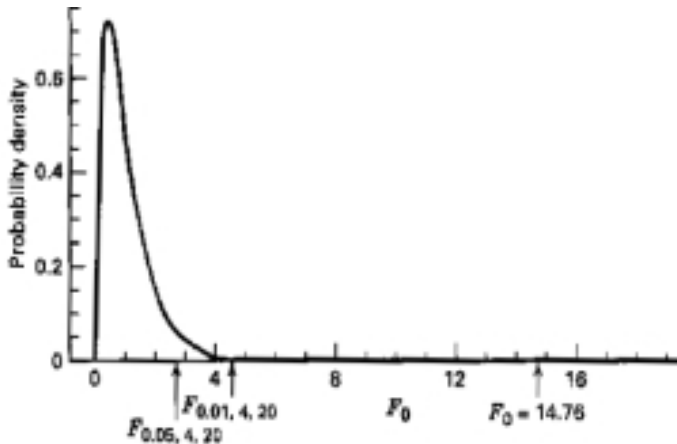
Figure 3-1 Box plots of tensile strength versus cotton weight percentage.

ANOVA Computer Output

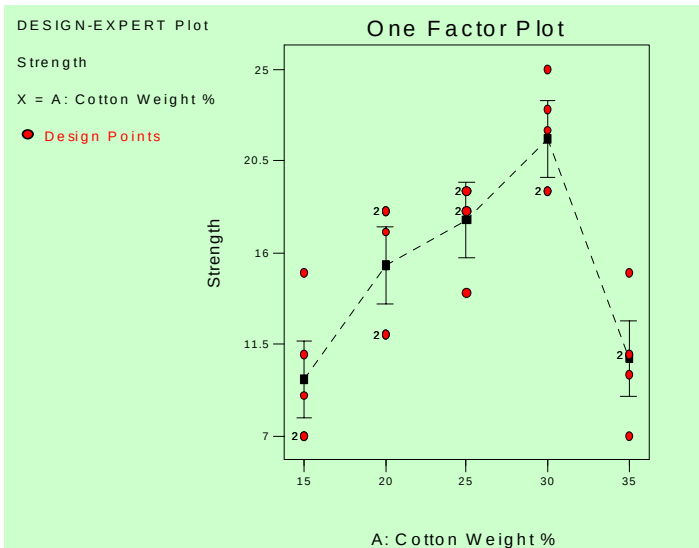
ANOVA						
<i>Source of Variation</i>	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>	<i>P-value</i>	<i>F crit</i>
Between Groups	475.76	4	118.94	14.75682	9.12794E-06	2.866081
Within Groups	161.2	20	8.06			
Total	636.96	24				

<i>Groups</i>	<i>Count</i>	<i>Sum</i>	<i>Average</i>	<i>Variance</i>
Row 1	5	49	9.8	11.2
Row 2	5	77	15.4	9.8
Row 3	5	88	17.6	4.3
Row 4	5	108	21.6	6.8
Row 5	5	54	10.8	8.2

ANOVA Computer Output



Graphical View of the Results

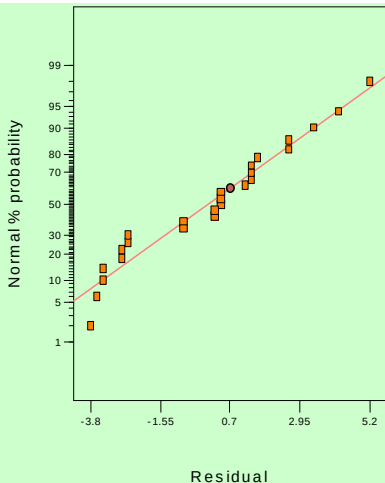


Model Adequacy Checking in the ANOVA

- Examination of **residuals**:

$$\begin{aligned}e_{ij} &= y_{ij} - \hat{y}_{ij} \\ &= y_{ij} - \bar{y}_i.\end{aligned}$$

- Residual plots** are very useful
- Normal probability plot** of residuals



Graphical Comparison of Means

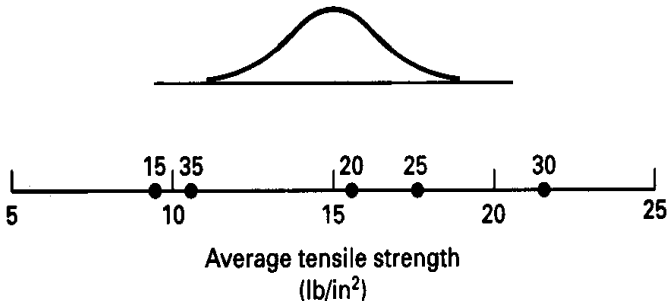


Figure 3-11 Tensile strength averages from the cotton weight percentage experiment in relation to a t distribution with a scale factor $\sqrt{MS_E/n} = \sqrt{8.06/5} = 1.27$.

Another Example

- Three machines are to be compared

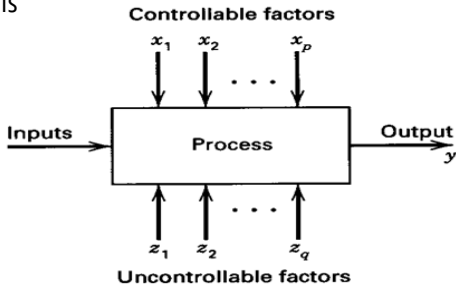
TABLE 10-1 Sample Outputs of Three Machines

	<i>Machine 1</i>	<i>Machine 2</i>	<i>Machine 3</i>	
	47	55	54	
	53	54	50	
	49	58	51	
	50	61	51	
	46	52	49	
$\bar{X}_i$	$\bar{X}_1 = 49$	$\bar{X}_2 = 56$	$\bar{X}_3 = 51$	$\bar{\bar{X}} = 52$
$(\bar{X}_i - \bar{\bar{X}})$	-3	4	-1	$\Sigma(\bar{X}_i - \bar{\bar{X}}) = 0\checkmark$
$(\bar{X}_i - \bar{\bar{X}})^2$	9	16	1	$\Sigma(\bar{X}_i - \bar{\bar{X}})^2 = 26$

- Are these machines truly different or not?

DOELEN VAN MODELLEREN

- Bepalen welke factoren x het meeste invloed hebben op de respons y
- Bepalen bij welke waarden x :
 - een bepaalde gewenste respons y bereikt wordt
 - de variabiliteit op y het kleinste is
 - de effecten van de oncontroleerbare factoren z minimaal zijn



DOELN VAN EXPERIMENTEREN

- Nieuwe processen ontwerpen tegen minimale kost
- Prestaties van bestaande processen verbeteren
- Betrouwbaarheid van bestaande processen verbeteren
- Robuustheid van bestaande processen verbeteren
- Alternatieven evalueren

TERMINOLOGIE

- Experiment
- Experimenteerpunt
- Experimentele run
- Replicaties
- Experimentele fout
- Experimenteergebied
- Proefschema (design)

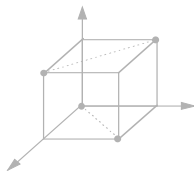
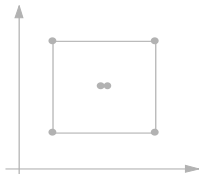
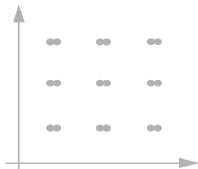
TERMINOLOGIE

Proefschema (design)

- Settings (levels) van factoren
- Replicaties voor welke experimenteerpunten
- Volgorde van de runs en replicaties
- Organisatorische maatregelen:
 - welke meetapparatuur?
 - wie doet de meting?
 - omstandigheden van de meting?
 - ...

ESSENTIËLE KENMERKEN VAN EEN GOED EXPERIMENTEEL ONTWERP

Het ontwerp is factorieel

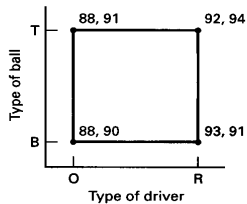


Omgaan met externe factoren:

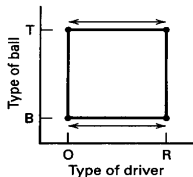
- Blocking (controleer het experiment)
- Randomising

(VOLLEDIG-)FACTORIEEL ONTWERP

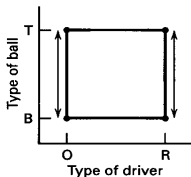
- In een factorieel ontwerp worden voor alle factoren een aantal levels vastgelegd.
- Vervolgens worden **alle** mogelijke runs getest, dwz. alle mogelijke combinaties van factor levels.
- Eenvoudigste vorm: het zogenaamde 2^k experimenteel ontwerp
k factoren telkens op 2 levels (hoog en laag)



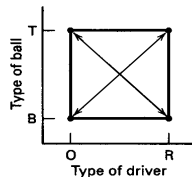
(a) Scores from the golf experiment



(b) Comparison of scores leading to the driver effect



(c) Comparison of scores leading to the ball effect



(d) Comparison of scores leading to the ball-driver interaction effect

Figure 1-5 Scores from the golf experiment in Figure 1-4 and calculation of the factor effects.

VOORDELEN VAN VOLLEDIG-FACTORIËLE EXPERIMENTEN

- Geen “confounding” van factoren
 - Laat eenduidige interpretatie van de resultaten toe
- Interactie tussen factoren is opspoorbaar, de grootte van de interactie-effecten kan geschat worden
- Garandeert maximale effectiviteit en nauwkeurigheid

EXTERNE FACTOREN (‘NUISANCE’ FACTORS)

	bekend	onbekend
controleerbaar	BLOCKING	
oncontroleerbaar	RANDOMISATION	

VOORBEELD BLOCKING

- Stel, 2 factoren **A** (levels a en A) en **B** (b of B), met 4 replicaties per experimenteerpunt.
- in totaal $2 \cdot 2 \cdot 4 = 16$ observaties.

AB, AB, AB, AB,
Ab, Ab, Ab, Ab,
aB, aB, aB, aB,
ab, ab, ab, ab

VOORBEELD BLOCKING

- Stel, 2 factoren **A** (levels a en A) en **B** (b of B), met 4 replicaties per experimenteerpunt.
- in totaal $2 \cdot 2 \cdot 4 = 16$ observaties.

Nieuwe blocking factor: weer

AB, AB, Ab, Ab, aB, aB, ab, ab	C	
AB, AB, Ab, Ab, aB, aB, ab, ab	c	

VOORBEELD BLOCKING

- Stel, 2 factoren **A** (levels a en A) en **B** (b of B), met 4 replicaties per experimenteerpunt.
- in totaal $2 \cdot 2 \cdot 4 = 16$ observaties.

ABC, ABC, AbC, AbC,
aBC, aBC, abC, abC



ABc, ABc, Abc, Abc,
aBc, aBc, abc, abc



VOORBEELD RANDOMISATIE

Nog beter is om te beschermen tegen onbekende en oncontroleerbare factoren door volgorde te randomiseren

abC, ABC, AbC, abC,
aBC, aBC, ABC, AbC

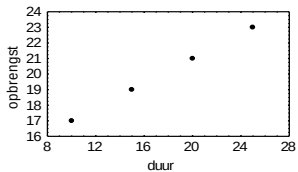
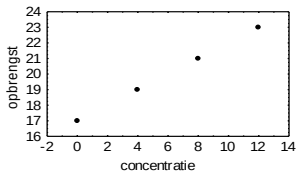
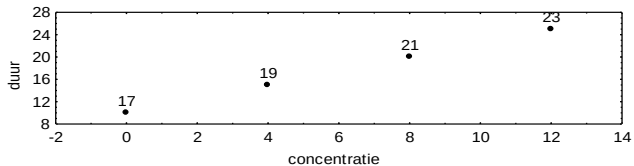


abc, aBc, Abc, Abc,
ABc, Abc, aBc, abc



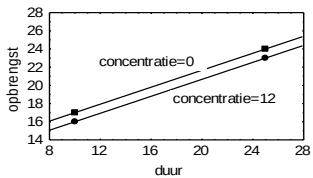
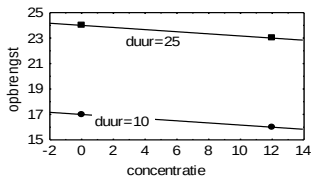
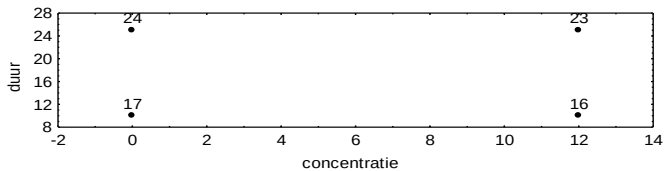
EEN VOORBEELD: OPBRENGST VAN EEN SCHEIKUNDIGE REACTIE

- Het vermoeden bestaat dat de opbrengst wordt beïnvloed door volgende factoren
 - **A: de concentratie van een additief** (0 % - 12 %);
 - **B: de reactieduur** (10 minuten - 25 minuten).
- Eén onderzoeker (Johan) denkt dat de concentratie een belangrijke positieve invloed heeft.
- Dit wordt betwist door een tweede onderzoeker (Nathalie) die stelt dat de reactieduur de belangrijkste factor is.
- Een experiment wordt opgezet met 4 nauwkeurig gemeten runs om dit te onderzoeken.



BESPREKING VOORBEELD

- We zitten hier duidelijk met een probleem. We stellen vast dat de opbrengst positief gecorreleerd is met zowel de concentratie als de duur, alleen zijn we niet in staat om uit te maken of beide correlaties ook betekenisvol zijn. Het enige dat we kunnen besluiten is dat er een gezamenlijk effect is. De reden is dat de instelwaarden voor beide factoren in het experiment zo (slecht) gekozen zijn, dat ze volledig “confounded” zijn, dus 100 % gecorreleerd.
- Daarom worden er in een tweede experiment nog twee runs uitgevoerd, die, samen met twee runs uit het eerste experiment, het volgend beeld opleveren:

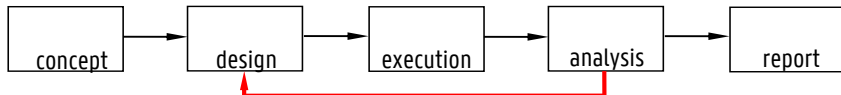


BESPREKING VOORBEELD

- Nu is de conclusie duidelijk : Nathalie heeft gelijk, de **duur van de reactie** heeft een positief effect op de opbrengst, de concentratie heeft daarentegen een licht negatief effect. De plots waaruit we dit duidelijk kunnen afleiden noemen we interactieplots.
- Beide experimenten bevatten 4 runs, en zijn dus even duur. Het eerste is waardeloos want het leert ons niet wat we willen weten. Het tweede experiment daarentegen levert nuttige informatie op. In het eerste experiment is de correlatie tussen beide factoren 1, in het tweede is de correlatie 0.

EEN EXPERIMENT PLANNEN, UITVOEREN EN ANALYSEREN

1. Probleemstelling en -formulering,
2. Keuze van factoren, levels en ranges,
3. Selectie van de responsvariabelen,
4. Ontwerp van het geschikte proefopzet,
5. Uitvoeren van het experiment,
6. (Statistische) analyse (ev. ontwerp aanpassen)
7. Conclusies en aanbevelingen



Factorial experimental design

Experimental units

An **experimental unit** (EU) is the item on/from which we obtain an **observation** (= data point)

- ▶ We assume: 1 observation per EU
- ▶ In case of multiple observations per EU: '**Repeated measures**' analysis

Experimental design:

- ▶ Every possible combination of factor levels is a **design point**
- ▶ A **run** is a design point for which observations are obtained
- ▶ Observations from the same run are called **replications**

The **design** specifies at least which factors, which runs and the number of replications per run ... but also *all* other specifications of **how the experiment is organised and conducted**

A **balanced** design contains n replications in all runs

Factorial Designs

- In a factorial experiment, **all possible combinations** of factor levels are tested.
- The golf experiment:
 - Type of driver
 - Type of ball
 - Walking versus riding
 - Type of beverage
 - Time of round
 - Weather
 - Type of golf spike

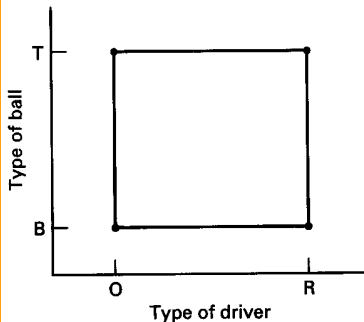
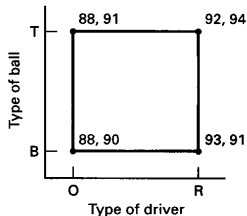
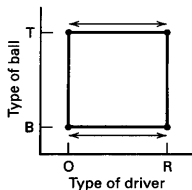


Figure 1-4 A two-factor factorial experiment involving type of driver and type of ball.

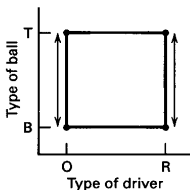
Factorial Design



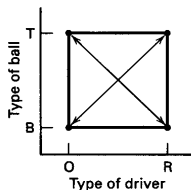
(a) Scores from the golf experiment



(b) Comparison of scores leading to the driver effect



(c) Comparison of scores leading to the ball effect



(d) Comparison of scores leading to the ball-driver interaction effect

Figure 1-5 Scores from the golf experiment in Figure 1-4 and calculation of the factor effects.

Factorial Designs (Several factors)

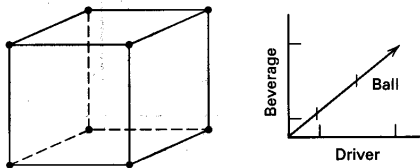


Figure 1-6 A three-factor factorial experiment involving type of driver, type of ball, and type of beverage.

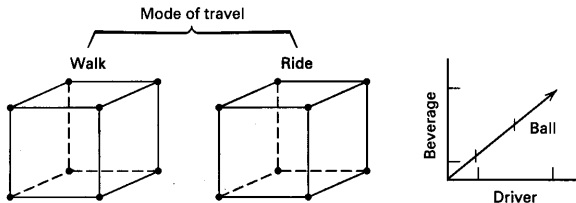


Figure 1-7 A four-factor factorial experiment involving type of driver, type of ball, type of beverage, and mode of travel.

Factorial Designs with Several Factors

A Fractional Factorial

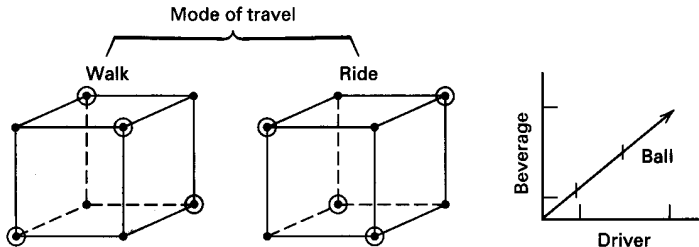


Figure 1-8 A four-factor fractional factorial experiment involving type of driver, type of ball, type of beverage, and mode of travel.

Nuisance factors: bad news

In the experiment, we want to study the effects of the chosen factors *only* ('internal' factors) and rule out the influence of all other outside factors ('external' or nuisance factors) as much as possible

- ▶ A nuisance factor is a factor that probably has some effect on the response, but is of no interest to the experimenter (aka. lurking or latent factor)
- ▶ Like all factors, it transmits variability to the response
→ undesirable, this variability needs to be minimised

Typical nuisance factors include batches of raw material, operators, pieces of test equipment, time (shifts, day of week), weather conditions, ...

Confounding

If the experimental design is such that the effect of a factor A can not be distinguished from that of factor B, factors A and B are said to be confounded

- ▶ A major challenge in experimental design is to prevent internal factors from being confounded with each other, and being confounded with nuisance factors

CODEREN VAN FACTOR LEVELS

Onderstel een 2^k ontwerp met:

- k ontwerpfactoren A, B, ...
- 2 levels per factor
- n replicaties (observaties) per experimenteerpunt

We **hercoderen** de levels altijd naar -1/+1: A = 80mm, A=130mm → A=-1, A=+1

- Komt neer op een translatie en herschaling
- Alle factoren krijgen zo **dezelfde 'schaal'**
- Interactiefactoren hebben op deze manier ook 2 levels -1/+1

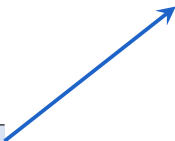
OPSTELLEN VAN DE ONTWERPMATRIX

'Stacked' vorm:

A	B	Y
-1	-1	5,94
-1	-1	5,50
-1	-1	6,10
-1	+1	5,50
-1	+1	5,25
-1	+1	5,16
+1	-1	4,16
+1	-1	6,72
+1	-1	2,72
+1	+1	6,25
+1	+1	5,90
+1	+1	4,97

'Unstacked' vorm:

A	B	Rep1	Rep2	Rep3
-1	-1	5,94	5,50	6,10
-1	+1	5,50	5,25	5,16
+1	-1	4,16	6,72	2,72
+1	+1	6,25	5,90	4,97



GROOTTE VAN EEN EFFECT

Hoe groot is het **effect** van factor Q op respons?

$$[Q] = (\text{gemiddelde respons bij } Q=+1) - (\text{gemiddelde respons bij } Q=-1)$$

Voorbeeld: hoofdeffect [A]

A	B	Rep1	Rep2	Rep3
-1	-1	5,94	5,50	6,10
-1	+1	5,50	5,25	5,16
+1	-1	4,16	6,72	2,72
+1	+1	6,25	5,90	4,97

gemiddelde respons bij A=+1 is

$$(4,16+6,72+2,72+6,25+5,90+4,97)/6 = 5,12$$

gemiddelde respons bij A=-1 is

$$(5,94+5,50+6,10+5,50+5,25+5,16)/6 = 5,575$$

$$[A] = 5,12 - 5,575 = -0,455$$

GROOTTE VAN EEN EFFECT

Hoe groot is het **effect** van factor Q op respons?

$$[Q] = (\text{gemiddelde respons bij } Q=+1) - (\text{gemiddelde respons bij } Q=-1)$$

Voorbeeld: hoofdeffect [B]

A	B	Rep1	Rep2	Rep3
-1	-1	5,94	5,50	6,10
-1	+1	5,50	5,25	5,16
+1	-1	4,16	6,72	2,72
+1	+1	6,25	5,90	4,97

gem. respons bij B=+1 is

$$(5,50+5,25+5,16+6,25+5,90+4,97)/6 = 5,505$$

gem. respons bij B=-1 is

$$(5,94+5,50+6,10+4,16+6,72+2,72)/6 = 5,19$$

$$[B] = 5,505 - 5,19 = 0,315$$

GROOTTE VAN EEN EFFECT

Hoe groot is het **effect** van factor Q op respons?

$$[Q] = (\text{gemiddelde respons bij } Q=+1) - (\text{gemiddelde respons bij } Q=-1)$$

Voorbeeld: **interactie-effect** [AB]

A	B	AB	Rep1	Rep2	Rep3
-1	-1	+1	5,94	5,50	6,10
-1	+1	-1	5,50	5,25	5,16
+1	-1	-1	4,16	6,72	2,72
+1	+1	+1	6,25	5,90	4,97

maak een nieuwe kolom AB
die het product is van kolommen A en B

$$[AB] = 5,78 - 4,92 = 0,857$$

2^k Factorial Design

The 2^2 Design

- 2^k is the notation used to indicate that a certain experimental design has k factors of interest, each at two levels.
- 2^2 design: Two factors A and B, each at two levels

	<u>A</u>	<u>B</u>
Low	-1	-1
High	+1	+1

- There are a total of four possible combinations.

2^k Factorial Design

The 2^2 Design

- The simplest design involves two factors A and B and n replicates.
- We are interested in the main effect of A, the main effect of B, and the interaction between A and B.
- Effects are calculated by:

**Average Response at high level - Average
Response at the low level.**

- A *large* effect would indicate a significant factor (or interaction). (How large is large?)
- **Contrasts** can be calculated and used to estimate the effects and then sums of squares.

Some Basic Definitions

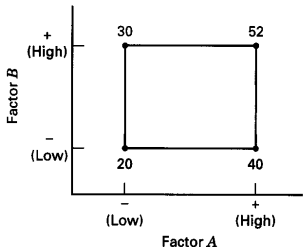


Figure 5-1 A two-factor factorial experiment, with the response (y) shown at the corners.

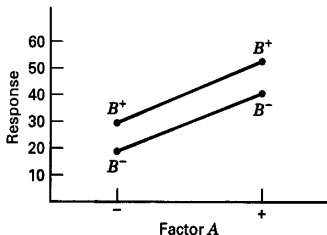


Figure 5-3 A factorial experiment without interaction.

Definition of a factor effect: The change in the mean response when the factor is changed from low to high.

$$A = \bar{y}_{A^+} - \bar{y}_{A^-} = \frac{40 + 52}{2} - \frac{20 + 30}{2} = 21$$

$$B = \bar{y}_{B^+} - \bar{y}_{B^-} = \frac{30 + 52}{2} - \frac{20 + 40}{2} = 11$$

$$AB = \frac{52 + 20}{2} - \frac{30 + 40}{2} = -1$$

The Case of Interaction:

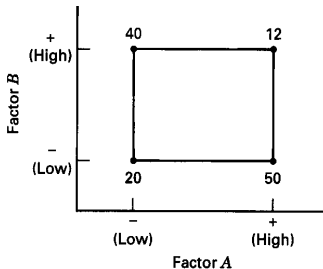


Figure 5-2 A two-factor factorial experiment with interaction.

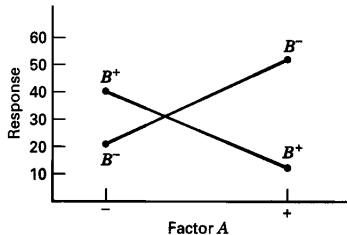


Figure 5-4 A factorial experiment with interaction.

$$A = \bar{y}_{A^+} - \bar{y}_{A^-} = \frac{50 + 12}{2} - \frac{20 + 40}{2} = 1$$

$$B = \bar{y}_{B^+} - \bar{y}_{B^-} = \frac{40 + 12}{2} - \frac{20 + 50}{2} = -9$$

$$AB = \frac{12 + 20}{2} - \frac{40 + 50}{2} = -29$$

2^k Factorial Design

The 2^2 Design

- Let the letters (1), a , b , and ab represent the totals of all n observations taken at these design points.
- Effect estimate of A:

$$\begin{aligned} A &= \bar{y}_{A^+} - \bar{y}_{A^-} \\ &= \frac{a + ab}{2n} - \frac{b + (1)}{2n} \\ &= \frac{1}{2n} [a + ab - b - (1)] \end{aligned}$$

2^k Factorial Design

The 2^2 Design

- Effect estimate of B:

$$\begin{aligned} B &= \bar{y}_{B^+} - \bar{y}_{B^-} \\ &= \frac{b + ab}{2n} - \frac{a + (1)}{2n} \\ &= \frac{1}{2n} [b + ab - a - (1)] \end{aligned}$$

- Effect estimate of AB:

$$\begin{aligned} AB &= \frac{ab + (1)}{2n} - \frac{a + b}{2n} \\ &= \frac{1}{2n} [ab + (1) - a - b] \end{aligned}$$

2^k Factorial Design

The 2^2 Design ($k = 2$):

- ▶ In the effects formulas on 55 and 56, the quantities in brackets are called **contrasts**. The contrasts for A, B, AB are

$$C_A = a + ab - b - (1) \quad C_B = ab + b - a - (1) \quad C_{AB} = ab + (1) - a - b$$

- ▶ The contrasts are used to calculate the sum of squares, for A, B and AB:

$$SS_A = \frac{C_A^2}{2^k n} = \frac{C_A^2}{4n} \quad SS_B = \frac{C_B^2}{2^k n} = \frac{C_B^2}{4n} \quad SS_{AB} = \frac{C_{AB}^2}{2^k n} = \frac{C_{AB}^2}{4n}$$

2^k Factorial Design

The 2^2 Design

- The sum of squares for A, B, and AB are:

$$SS_A = \frac{[a + ab - b - (1)]^2}{4n}$$
$$SS_B = \frac{[b + ab - a - (1)]^2}{4n}$$
$$SS_{AB} = \frac{[ab + (1) - a - b]^2}{4n}$$

Chemical Process Example

Factor		Treatment Combination	Replicate			Total
A	B		I	II	III	
-	-	A low, B low	28	25	27	80
+	-	A high, B low	36	32	32	100
-	+	A low, B high	18	19	23	60
+	+	A high, B high	31	30	29	90

A = reactant concentration,

B = catalyst amount, and

The response variable y = recovery

Chemical Process Example

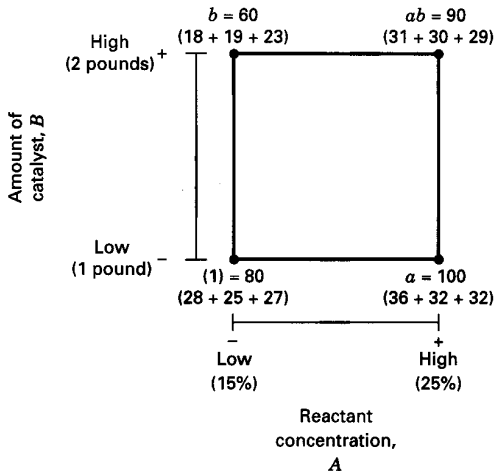


Figure 6-1 Treatment combinations in the 2^2 design.

“-” and “+” denote the low and high levels of a factor, respectively.

Geometrically, the four runs represent the corners of a square.

Factors can be quantitative or qualitative, although their treatment in the final model will be different.

The effect estimates are:

$$A = 8.33,$$

$$B = -5.00,$$

$$AB = 1.67$$

Chemical Process Example

- The ANOVA table for the router example is then:

Response: Conversion

ANOVA for Selected Factorial Model

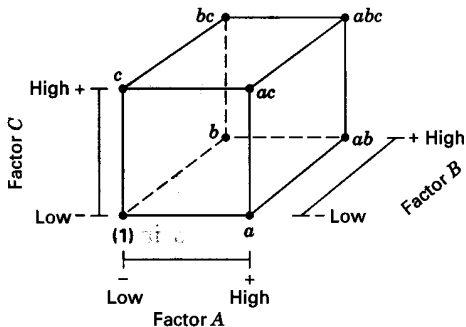
Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F
Model	291.67	3	97.22	24.82	0.0002
A	208.33	1	208.33	53.19	< 0.0001
B	75.00	1	75.00	19.15	0.0024
AB	8.33	1	8.33	2.13	0.1828
Pure Error	31.33	8	3.92		
Cor Total	323.00	11			

Std. Dev.	1.98	R-Squared	0.9030
Mean	27.50	Adj R-Squared	0.8666
C.V.	7.20	Pred R-Squared	0.7817
PRESS	70.50	Adeq Precision	11.669

- The F -test for the “model” source is testing the significance of the overall model; that is, is either A , B , or AB or some combination of these effects important?

The 2^3 Factorial Design



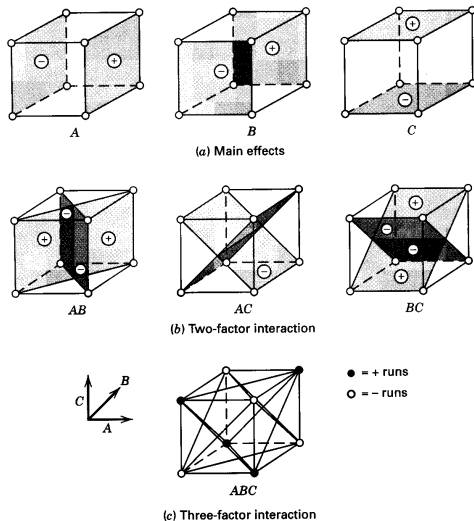
(a) Geometric view

Run	Factor		
	A	B	C
1	-	-	-
2	+	-	-
3	-	+	-
4	+	+	-
5	-	-	+
6	+	-	+
7	-	+	+
8	+	+	+

(b) The design matrix

Figure 6-4 The 2^3 factorial design.

Effects in The 2^3 Factorial Design



$$A = \bar{y}_{A^+} - \bar{y}_{A^-}$$

$$B = \bar{y}_{B^+} - \bar{y}_{B^-}$$

$$C = \bar{y}_{C^+} - \bar{y}_{C^-}$$

etc, etc, ...

Figure 6-5 Geometric presentation of contrasts corresponding to the main effects and interactions in the 2^3 design.

2^k Factorial Design

The 2^k Design for $k \geq 3$ Factors

Effect Estimate for A:

$$A = \bar{y}_{A^+} - \bar{y}_{A^-} = \frac{1}{4n} [a + ab + ac + abc - b - c - bc - (1)]$$

Effect Estimate for B:

$$B = \bar{y}_{B^+} - \bar{y}_{B^-} = \frac{1}{4n} [b + ab + bc + abc - a - c - ac - (1)]$$

Effect Estimate for C:

$$C = \bar{y}_{C^+} - \bar{y}_{C^-} = \frac{1}{4n} [c + ac + bc + abc - a - b - ab - (1)]$$

2^k Factorial Design

The 2^k Design for $k \geq 3$ Factors

Effect Estimate for AB:

$$AB = \bar{y}_{AB^+} - \bar{y}_{AB^-} = \frac{1}{4n} [ab + (1) + abc + c - b - a - bc - ac]$$

Effect Estimate for AC:

$$AC = \bar{y}_{AC^+} - \bar{y}_{AC^-} = \frac{1}{4n} [ac + (1) + abc + b - a - c - ab - bc]$$

Effect Estimate for BC:

$$BC = \bar{y}_{BC^+} - \bar{y}_{BC^-} = \frac{1}{4n} [bc + (1) + abc + a - b - c - ab - ac]$$

2^k Factorial Design

The 2^k Design for $k \geq 3$ Factors

Effect Estimate for ABC:

$$ABC = \bar{y}_{ABC^+} - \bar{y}_{ABC^-} = \frac{1}{4n} [abc - bc - ac + c - ab + b + a - (1)]$$

In general, the effects can be estimated using:

$$Effect = \frac{Contrast}{n2^{k-1}}$$

The sum of squares for any effect is:

$$SS = \frac{(Contrast)^2}{n2^k}$$

2^k Factorial Design

Other Methods for Judging the Significance of Effects

- The standard error of any effect estimate in a 2^k design is:

$$s.e.(Effect) = \sqrt{\frac{\hat{\sigma}^2}{n2^{k-2}}}$$

- Two standard deviation limits on any estimated effect is:

$$Effect\ estimate \pm 2[s.e.(effect)]$$

2^k Factorial Design

Other Methods for Judging the Significance of Effects

$$\textit{Effect estimate} \pm 2[\textit{s.e.}(\textit{effect})]$$

- This interval is an approximate 95% confidence interval on the estimated effect.
- If zero is contained within the 95% confidence interval, then that effect is essentially zero and the corresponding factor is not significant at the $\alpha = 0.05$ level.

2^k Factorial Design

Example

- An experiment was performed to investigate the surface finish of a metal part. The experiment is a 2^3 factorial design in the factors feed rate (A), depth of cut (B), and tool angle (C), with $n = 2$ replicates.

2^k Factorial Design

Example

		Design Factors			Surface Finish	Totals
Run		A	B	C		
1	(1)	-1	-1	-1	9, 7	16
2	a	1	-1	-1	10, 12	22
3	b	-1	1	-1	9, 11	20
4	ab	1	1	-1	12, 15	27
5	c	-1	-1	1	11, 10	21
6	ac	1	-1	1	10, 13	23
7	bc	-1	1	1	10, 8	18
8	abc	1	1	1	16, 14	30

The larger the value, the rougher the surface finish.

2^k Factorial Design

Example – Analysis of Variance Table

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0	P-value
A	45.5625	1	45.5625	18.69	2.54×10^{-3}
B	10.5625	1	10.5625	4.33	0.07
C	3.0625	1	3.0625	1.26	0.29
AB	7.5625	1	7.5625	3.10	0.12
AC	0.0625	1	0.0625	0.03	0.88
BC	1.5625	1	1.5625	0.64	0.45
ABC	5.5625	1	5.5625	2.08	0.19
Error	19.5000	8	2.4375		
Total	92.9375	15			

2^k Factorial Design

A Single Replicate of the 2^k Design

- As the number of factors in a factorial experiment increases, the number of effects that can be estimated also increases.
- In most situations, the **sparsity of effects principle** applies.
- For a large number of factors, say $k > 5$, it is common practice to run only a single replicate of the 2^k design and pool or combine the higher-order interactions as estimate of error.

2^k Factorial Design

Example

- Nitride etch process on a single-wafer plasma etcher. There are four factors of interest. The response is etch rate for silicon nitride. A single replicate is used.

	Gap	Pressure	C ₂ F ₆ Flow	Power
Level	A (cm)	B (m Torr)	C (SCCM)	D (W)
Low (-)	0.80	450	125	275
High (+)	1.20	550	200	325

2^k Factorial Design

Example

Run	A (Gap)	B (Pressure)	C (C ₂ F ₆ Flow)	D (Power)	Etch Rate (? /min)
1	-1	-1	-1	-1	550
2	1	-1	-1	-1	669
3	-1	1	-1	-1	604
4	1	1	-1	-1	650
5	-1	-1	1	-1	633
6	1	-1	1	-1	642
7	-1	1	1	-1	601
8	1	1	1	-1	635
9	-1	-1	-1	1	1037
10	1	-1	-1	1	749
11	-1	1	-1	1	1052
12	1	1	-1	1	868
13	-1	-1	1	1	1075
14	1	-1	1	1	860
15	-1	1	1	1	1063
16	1	1	1	1	729

2^k Factorial Design

Example

- The estimated effects are found to be

A = -101.625	AD = -153.625
B = -1.625	BD = -0.625
AB = -7.875	ABD = 4.125
C = 7.375	CD = -2.125
AC = -24.875	ACD = 5.625
BC = -43.875	BCD = -25.375
ABC = -15.625	ABCD = -40.125
D = 306.125	

Nitride Etch example

The 'full model' will use up all degrees of freedom (df), so none are left for inference. This full model is said to be **saturated**.

```
anova(lm(Y~A*B*C*D,data=df))
# Warning in anova.lm(lm(Y ~ A * B * C * D, data = df)): ANOVA F-tests on an essentially
perfect fit are unreliable
# Analysis of Variance Table
#
# Response: Y
#
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
# A	1	41311	41311	NaN	NaN
# B	1	11	11	NaN	NaN
# C	1	218	218	NaN	NaN
# D	1	374850	374850	NaN	NaN
# A:B	1	248	248	NaN	NaN
# A:C	1	2475	2475	NaN	NaN
# B:C	1	7700	7700	NaN	NaN
# A:D	1	94403	94403	NaN	NaN
# B:D	1	2	2	NaN	NaN
# C:D	1	18	18	NaN	NaN
# A:B:C	1	977	977	NaN	NaN
# A:B:D	1	68	68	NaN	NaN
# A:C:D	1	127	127	NaN	NaN
# B:C:D	1	2576	2576	NaN	NaN
# A:B:C:D	1	6440	6440	NaN	NaN
# Residuals	0	0	NaN		

Nitride Etch example

That is why we 'pool higher-order interactions' into the error term:

```
anova(lm(Y~(A+B+C+D)^2,data=df))
# Analysis of Variance Table
#
# Response: Y
#
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
# A	1	41311	41311	20.2765	0.006382	**
# B	1	11	11	0.0052	0.945391	
# C	1	218	218	0.1068	0.757069	
# D	1	374850	374850	183.9879	3.903e-05	***
# A:B	1	248	248	0.1218	0.741351	
# A:C	1	2475	2475	1.2148	0.320582	
# A:D	1	94403	94403	46.3357	0.001042	**
# B:C	1	7700	7700	3.7794	0.109498	
# B:D	1	2	2	0.0008	0.978978	
# C:D	1	18	18	0.0089	0.928641	
# Residuals	5	10187	2037			
# ---						
# Signif. codes:	0	'***'	0.001	'**'	0.01	'*' 0.05 '.' 0.1 ' ' 1

Fractional Replication of the 2^k Design

- As the number of factors in a 2^k increase, the number of runs required increases rapidly.
- If we can assume that some higher-order interactions are negligible, then a **fractional factorial design** can be used to gain information on main effects and low-order interactions.

Fractional Replication of the 2^k Design

The One-Half Fraction of the 2^k

- Contains 2^{k-1} runs,
- Often called a 2^{k-1} fractional factorial design,
- Consider a three-factor design: 2^{3-1} that is one half of a full 2^3

Fractional replication of the 2^k design

The one-half fraction of the 2^k

- ▶ Table for 2^3 with all main effects and interactions, including the 'identity' column I

Run	Factorial effect							
	I	A	B	C	AB	AC	BC	ABC
a	+	+	-	-	-	-	+	+
b	+	-	+	-	-	+	-	+
c	+	-	-	+	+	-	-	+
abc	+	+	+	+	+	+	+	+
ab	+	+	+	-	+	-	-	-
ac	+	+	-	+	-	+	-	-
bc	+	-	+	+	-	-	+	-
(1)	+	-	-	-	+	+	+	-

- ▶ Suppose runs a , b , c and abc are chosen as the one-half fraction i.e. the 4 runs shown in the top half of the table
- ▶ Note that these selected runs all have '+' on the ABC effect, so we can characterise this top-half by the relation $I = ABC$
→ all selected runs satisfy this relation

Fractional Replication of the 2^k Design

The One-Half Fraction of the 2^k

- The identity element I is also plus for the four runs, therefore $I = ABC$, which is the **defining relation** for the design.
- The defining relation can be used to find **aliases** of the main effects and interactions.
- To find the aliases of a main effect or interaction, multiply the entire defining relation by that term. (*Note: any column multiplied by itself always results in the identity column, i.e. $A \cdot A = I$*)

Fractional Replication of the 2^k Design

The One-Half Fraction of the 2^k

- With defining relation, $I = ABC$, some of the *aliases* are:

- The alias of main effect A is:

$$A \cdot I = A \cdot ABC = BC$$

That is, A is aliased with (identical to) the BC interaction.

- The alias of the main effect B is:

$$B \cdot I = B \cdot ABC = AC$$

That is, B is aliased with the AC interaction.

- The alias of the AB interaction is:

$$AB \cdot I = AB \cdot ABC = C$$

That is, the AB interaction is aliased with the main effect C.

Fractional Replication of the 2^k Design

The One-Half Fraction of the 2^k

- **Projection of the 2^{k-1} Design**
 - If one or more factors from a one-half fraction of a 2^k can be dropped, the design will **project** into a full factorial design.
 - Projection is highly useful in screening experiments.

Fractional Replication of the 2^k Design

The One-Half Fraction of the 2^k

- **Design Resolution**

- Design resolution is useful in categorizing designs. The resolution of a design is usually denoted by Roman numerals.
- Of particular importance are designs of Resolution III, IV, and V.
- The resolution of a design indicates the alias relationship among the factors.

Fractional Replication of the 2^k Design

The One-Half Fraction of the 2^k

- **Design Resolution**
 - **Resolution III designs.** No main effects are aliased with one another. Main effects are aliased with two factor interactions.
 - **Resolution IV designs.** No main effect is aliased with any other main effect or two-factor interaction. Two-factor interactions are aliased with each other.
 - **Resolution V designs.** No main effects are aliased with any other main effect or two-factor interactions. Two-factor interactions are not aliased with one another. Two-factor interactions are aliased with three-factor interactions.

Fractional Replication of the 2^k Design

The One-Half Fraction of the 2^k

- Design Resolution
 - For example, the notation 2^{3-1}_{III} indicates that the design is a one-half fraction of a 2^3 with resolution III. From this, we know that main effects are not aliased with one another, but are aliased with two factor interactions.

Fractional replication of the 2^k design

Smaller fractions: the 2^{k-p} fractional factorial design

- ▶ While the half-fraction is useful, it may be necessary to use even **smaller** fractions of the 2^k
- ▶ Consider a design with $k = 11$ factors

→ a full 2^k would require 2048 runs

→ a half-fraction, 2^{k-1} would **still** require 1024 runs!

- ▶ How about a design for all 11 factors that would only require 32 runs?

Some assumptions would have to be made, but this is a possible design

11 factor in 32 runs (2^5) would be a fraction $2^{-6} = \frac{1}{2^6} = \frac{1}{64}$ of the full factorial

So in general, for a screening of k factors (each with 2 levels), using only 2^r runs, you require a $2^{-k+r} = 2^{-p}$ fraction of the full factorial 2^k

Fractional Replication of the 2^k Design

Smaller Fractions: The 2^{k-p} Fractional Factorial Design

- Setting up a 2^{k-p} design requires:
 1. Setting up a full factorial design for $k-p$ factors.
 2. Generate the remaining p columns by selecting appropriate **design generators**.

Fractional Replication of the 2^k Design

To illustrate, consider a 2^{6-2} fractional factorial design:

1. Set up a full factorial design for $6 - 2 = 4$ factors:

Run	A	B	C	D
1	-	-	-	-
2	+	-	-	-
3	-	+	-	-
4	+	+	-	-
5	-	-	+	-
6	+	-	+	-
7	-	+	+	-
8	+	+	+	-
9	-	-	-	+
10	+	-	-	+
11	-	+	-	+
12	+	+	-	+
13	-	-	+	+
14	+	-	+	+
15	-	+	+	+
16	+	+	+	+

Fractional Replication of the 2^k Design

1. Generate the remaining 2 columns using a design generator that involves the 4 factors used in part 1. For example, the column for factor E could be generated using ABC. The column for factor F could be generated using BCD.

Run	A	B	C	D	E	F
1	-	-	-	-	-	-
2	+	-	-	-	+	-
3	-	+	-	-	+	+
4	+	+	-	-	-	+
5	-	-	+	-	+	+
6	+	-	+	-	-	+
7	-	+	+	-	-	-
8	+	+	+	-	+	-
9	-	-	-	+	-	+
10	+	-	-	+	+	+
11	-	+	-	+	+	-
12	+	+	-	+	-	-
13	-	-	+	+	+	-
14	+	-	+	+	-	-
15	-	+	+	+	-	+
16	+	+	+	+	+	+

Fractional Replication of the 2^k Design

- In this example there are **two design generators**:
 - $E = ABC \rightarrow I = ABCE$
 - $F = BCD \rightarrow I = BCDF$
- To find the **complete defining relation**, multiply all pairs of the design generators (in this case there are only two so we would multiply $(ABCE)(BCDF) = ADEF$)
- Therefore, the **complete defining relation** from which all aliases can be found for main effects and interactions is:

$$\underline{I = ABCE = BCDF = ADEF}$$

- By definition, the length of the smallest “**word**” in the defining relation is also the *resolution* of the design. In this case, the resolution is IV.

Fractional Replication of the 2^k Design

Smaller Fractions: The 2^{k-p} Fractional Factorial Design

- Selection of Design Generators:
 - Design generators should not be chosen arbitrarily.
 - Choosing the incorrect design generator can result in a design of smaller resolution than possible.
 - Select the design generators that will result in the highest possible design resolution.

■ TABLE 13.23

Selected 2^{k-p} Fractional Factorial Designs (from *Design and Analysis of Experiments*, 6th ed., by D. C. Montgomery, John Wiley, 2005)

Number of Factors, k	Fraction	Number of Runs	Design Generators	Number of Factors, k	Fraction	Number of Runs	Design Generators		
3	2_{III}^{3-1}	4	$C = \pm AB$	10	2_{IV}^{9-3}	64	$J = \pm BCEFG$		
4	2_{IV}^{4-1}	8	$D = \pm ABC$				$G = \pm ABCD$		
5	2_V^{5-1}	16	$E = \pm ABCD$				$H = \pm ACEF$		
	2_{III}^{5-2}	8	$D = \pm AB$				$J = \pm CDEF$		
6	2_{VI}^{6-1}	32	$E = \pm AC$		2_{IV}^{9-4}	32	$F = \pm BCDE$		
			$F = \pm ABCDE$				$G = \pm ACDE$		
	2_{IV}^{6-2}	16	$E = \pm ABC$				$H = \pm ABDE$		
			$F = \pm BCD$				$J = \pm ABCE$		
	2_{III}^{6-3}	8	$D = \pm AB$		2_{III}^{9-5}	16	$E = \pm ABC$		
			$E = \pm AC$				$F = \pm BCD$		
7	2_{VII}^{7-1}	64	$F = \pm BC$				$G = \pm ACD$		
			$G = \pm ABCDEF$				$H = \pm ABD$		
			$F = \pm ABCD$		$J = \pm ABCD$				
			$G = \pm ABDE$		$H = \pm ABCG$				
	2_{IV}^{7-2}	32	$E = \pm ABC$		$J = \pm BCDE$				
			$F = \pm BCD$		$K = \pm ACDF$				
	2_{III}^{7-3}	16	$G = \pm ACD$		2_{IV}^{10-4}	64	$G = \pm BCDF$		
			$D = \pm AB$				$H = \pm ACDF$		
8	2_{VIII}^{7-4}	8	$E = \pm AC$				2_{IV}^{10-5}	32	$J = \pm ABDE$
			$F = \pm BC$						$K = \pm ABCE$
	2_V^{8-2}	64	$G = \pm ABC$		$F = \pm ABCD$				
			$G = \pm ABCD$		$G = \pm ABCE$				
			$H = \pm ABEF$		$H = \pm ABDE$				
			$F = \pm ABC$		$J = \pm ACDE$				
2_{IV}^{8-3}	32	$G = \pm ABD$	2_{III}^{10-6}		16	$K = \pm BCDE$			
		$H = \pm BCDE$				$E = \pm ABC$			
		$E = \pm BCD$				$F = \pm BCD$			
		$F = \pm ACD$				$G = \pm ACD$			
9	2_{IV}^{8-4}	16				$G = \pm ABC$	$H = \pm ABD$		
						$H = \pm ABD$	$J = \pm ABCD$		
						$H = \pm ABD$	$K = \pm AB$		
						$H = \pm ACD$			
	2_{VI}^{9-2}	128				$H = \pm ACD$			
			$H = \pm ACD$						

References

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